



Atlas Model  
**ABP 180 ASPHALT  
BATCH MIX PLANT**

With **Computerised Panel bag house**  
type pollution control unit -rated  
capacity **180 tph @ 3%**  
**moisture content.**



## SCOPE OF SUPPLY

| Sr. No. | Description                      | Page No. |
|---------|----------------------------------|----------|
| 1       | General Description              | 01       |
| 2       | Cold Bin Feeder                  | 02       |
| 3       | Slinger conveyor                 | 03       |
| 4       | Dryer Drum                       | 04       |
| 5       | Burner Modulating                | 06       |
| 6       | Bag House Filter Unit            | 07       |
| 7       | Fined & re-cycling System        | 08       |
| 8       | Hot Elevator                     | 09       |
| 9       | Vibrator Screen                  | 10       |
| 10      | Hot Aggregate Bin                | 11       |
| 11      | Weighing system                  | 12       |
| 12      | Pug Mill type Mixer Unit         | 13       |
| 13      | Fines & Filler Feeding system    | 14       |
| 14      | Asphalt Pumps                    | 15       |
| 16      | Bitumen Storage & Feeding System | 16       |
| 17      | Control cabin                    | 18       |
| 18      | Operational Accessories &        | 20       |
| 19      | Other Ancillary Items            | 21       |

### General Description

**Stationary, Batch type, Fully Automatic Asphalt Mixing Plant – Model ABP 180.**

| Specification Category               | Details                                                                                        |
|--------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Production and Capacity</b>       | 180 tons per hour (based on 1.6 ton/m <sup>3</sup> density and 150°C output temperature).      |
|                                      | 2250 Kg batch with 80 cycles per hour.                                                         |
|                                      | Rated capacity considers 3% moisture content in aggregates.                                    |
|                                      | Maintain mix temperature with ±5°C precision.                                                  |
| <b>Material Handling and Quality</b> | Aggregates must be clean, free from dirt, and without oversized material to prevent blockages. |
|                                      | Less than 3% moisture, free from lumps or agglomeration.                                       |
|                                      | Accurate dosing with preheating for consistency.                                               |
|                                      | Advanced bag filters or scrubbers to meet environmental norms.                                 |
| <b>Automation and Control</b>        | Fully automatic electro-pneumatic operation with manual override.                              |

|                                    |                                                                               |
|------------------------------------|-------------------------------------------------------------------------------|
|                                    | Integrated systems for real-time data acquisition and process control.        |
|                                    | Emergency stop switches and fire suppression systems.                         |
|                                    | Energy-efficient burner systems to optimize fuel usage.                       |
| <b>Mixing Homogeneity</b>          | High-performance twin-shaft or planetary mixers for uniform output.           |
| <b>Maintenance and Scalability</b> | Modular design for quick servicing and centralized lubrication systems.       |
|                                    | Use of wear-resistant materials in critical areas such as mixer linings.      |
|                                    | Provision for capacity upgrades or integration with advanced technologies.    |
| <b>Environmental Compliance</b>    | Compliance with <b>Euro, EPA</b> , or equivalent environmental standards.     |
|                                    | Capability to incorporate RAP for sustainable production.                     |
| <b>Operational Versatility</b>     | Ability to produce mixes of varying grades (e.g., dense graded, open graded). |
|                                    | Stationary or mobile design options based on project requirements.            |

### Cold Bin Feeder

| Component                     | Specification                                                                                     |
|-------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Cold Aggregate Hoppers</b> | Five hoppers with a heaped capacity of approx. 40 m <sup>3</sup> (10 m <sup>3</sup> each).        |
|                               | Wide opening bin width to accommodate various loader capacities.                                  |
|                               | Anti-bridging bin design for smooth and consistent material flow, irrespective of material level. |
|                               | Sturdy and proven construction for durability.                                                    |
| <b>Belt Feeders</b>           | Four feeders, each 500 mm wide and 2.750 m long, with adjustable radial discharge gates.          |
|                               | Adjustable skirts to prevent leakage between bin bottom and belt surface.                         |
|                               | Feeding capacity: 50 Tons per Hour per hopper.                                                    |
|                               | Powered by 1.5 kW (2 HP) electric motors for each feeder (total: 5 x 1.5 kW).                     |
| <b>Feeding Units</b>          | Components include:                                                                               |
|                               | Head and tail pulleys.                                                                            |
|                               | Single roll idlers.                                                                               |
|                               | Endless rubber conveyor.                                                                          |
|                               | Screw-type belt tensioners.                                                                       |

|                          |                                                                                                                                                                           |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | Replaceable rubber skirting.                                                                                                                                              |
|                          | No-flow indicators for monitoring.                                                                                                                                        |
|                          | Vibrator on sand bin (0.3 HP or 0.2 kW) for enhanced material flow.<br>- Low height feeder due to which minimal ramp making therefore easy maintenance and less expenses. |
| <b>Power and Control</b> | Static inverter variable speed controls with transmission via gear reduction unit.                                                                                        |



| Parameter                        | Details                                                                     |
|----------------------------------|-----------------------------------------------------------------------------|
| <b>Type</b>                      | Feeder frame integrated, endless conveyor.                                  |
| <b>Belt Width</b>                | 600 mm.                                                                     |
| <b>Length</b>                    | 32 meters.                                                                  |
| <b>Rated Feeding Capacity</b>    | 200 Tons per Hour.                                                          |
| <b>Motor Type</b>                | 7.5 HP (5.5 Kw) squirrel cage electric motor.                               |
| <b>Drive System</b>              | Speed reduction unit.                                                       |
| <b>Idlers</b>                    | Trough idlers, roller guides, and return rollers for smooth belt operation. |
| <b>Belt Tensioning Mechanism</b> | Screw-type belt tensioners for precise tension adjustment.                  |
| <b>Connected Power</b>           | 7.55 HP (5.5 Kw)                                                            |

|                              |                                                         |
|------------------------------|---------------------------------------------------------|
| <b>Construction Material</b> | Heavy-duty construction for durability and reliability. |
|------------------------------|---------------------------------------------------------|

### Slinger conveyor

| Component                     | Specification                                                       |
|-------------------------------|---------------------------------------------------------------------|
| <b>Type</b>                   | High-speed slinger (throwing) conveyor.                             |
| <b>Belt Dimensions</b>        | 600 mm wide x 18 m long.                                            |
| <b>Rated Feeding Capacity</b> | 200 Tons per Hour (Tph).                                            |
| <b>Power</b>                  | 7.5 HP (5.5 Kw) squirrel cage electric motor.                       |
| <b>Drive Mechanism</b>        | Speed reduction unit for controlled and efficient operation.        |
| <b>Features</b>               | - Equipped with trough idlers for optimal belt support.             |
|                               | - Roller guides for proper belt alignment.                          |
|                               | - Return rollers to ensure smooth belt return movement.             |
|                               | - Mounted on a rigid structural frame for stability and durability. |

### Dryer Drum

| Component                 | Specification                                                                                                     |
|---------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Type</b>               | Cylindrical, inclined Friction Wheel Drive counter flow type dryer drum.                                          |
| <b>Dimensions</b>         | - <b>Drum Diameter:</b> 2,100 mm (2.1 m).                                                                         |
|                           | - <b>Drum Length:</b> 8,000 mm (8.0 m).                                                                           |
|                           | - <b>Drum Slope:</b> 5 degrees for effective material flow.                                                       |
| <b>Temperature</b>        | Maximum output temperature: 300°C at the drum outlet.                                                             |
| <b>Drying Capacity</b>    | Internal flights are designed to provide higher drying capacity, even with changes in aggregate moisture content. |
| <b>Heating Efficiency</b> | Internal dryer flights are field-proven and specifically arranged for:                                            |
|                           | - Enhanced heating efficiency.                                                                                    |

|                            |                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------|
|                            | - Homogeneous heating of aggregates with varying grain sizes.                           |
| <b>Support System</b>      | - Four heavy-duty trunnion rollers for smooth drum rotation.                            |
|                            | - Two thrust rollers for axial stability.                                               |
|                            | - Two trunnion rings with friction wheel drive for consistent operation.                |
|                            | - Rigid frame construction for durability and stability.                                |
| <b>Power</b>               | 4 x 15 HP (10 Kw) motors for friction wheel drive.                                      |
| <b>Bearings</b>            | Spherical roller type FAG/SKF for durability and load distribution.                     |
| <b>Additional Features</b> | - Hot aggregate temperature measured via a thermocouple-type probe at the dryer outlet. |



## Burner

| Component            | Specification             |
|----------------------|---------------------------|
| <b>Model</b>         | FCMB/LO/M/1000            |
| <b>Type</b>          | Air Automize / Modulating |
| <b>Capacity</b>      | 14 Mw                     |
| <b>Fuel Type</b>     | Diesel / LDO              |
| <b>Burner Rating</b> | 295 to 1180 Kg per hour   |
| <b>Blower</b>        | - Power: 30 HP (22 Kw)    |
| <b>Fuel Pump</b>     | - Power: 7.5 HP (5.5 Kw)  |

|                |                                                                |
|----------------|----------------------------------------------------------------|
| <b>Control</b> | Automatic on/off control for efficient and reliable operation. |
|----------------|----------------------------------------------------------------|

## Two Stage Dry Dust Collection System (Bag House Filter Unit)

### Primary Dust Collecting Unit

| Component                             | Specification                                                                       |
|---------------------------------------|-------------------------------------------------------------------------------------|
| <b>Primary Dust Collection System</b> | Cyclone or pre-separator to trap larger dust particles from the dryer drum exhaust. |
|                                       | Includes scavenging for the batching tower by screw conveyor.                       |
| <b>Type</b>                           | Screw conveyor IVA Make for dust transfer.                                          |
| <b>Conveyor Capacity</b>              | 10 HP (7.5 Kw)                                                                      |
| <b>High-Capacity Exhaust Fan</b>      | Turbo fan with Variable Frequency Drive (VFD) for enhanced control.                 |
| <b>Fan Power</b>                      | 120 HP (90 Kw)                                                                      |

### Bag House Filter Unit



| Component                          | Specification                                                                                                    |
|------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Dust Collection System Type</b> | Reverse Air Flow type bag filter.                                                                                |
| <b>Number of Bags</b>              | 384 bags (304 mm x 2000 mm).                                                                                     |
| <b>Filter Area</b>                 | 510 m <sup>2</sup> (approx.).                                                                                    |
| <b>Bag Material</b>                | Virgin Nomex type bags for high-efficiency filtration.                                                           |
| <b>Normal Temperature Limit</b>    | 180°C for optimal performance.                                                                                   |
| <b>Dust Emission</b>               | 0.1 g/m <sup>3</sup> (Normal operation).                                                                         |
| <b>De-dusting System</b>           | Fully automatic pneumatic type minimize maintenance of air compressors, solenoid valves, and pneumatic controls. |
| <b>Safety Features</b>             | Safety interlocks (e.g., high exhaust temperature) to ensure safe operation.                                     |
| <b>Connected Power</b>             | 16 Pneumatic Cylinder Control by Solenoid Valve                                                                  |
| <b>Capacity</b>                    | 45,000 m <sup>3</sup> /hr air flow capacity.                                                                     |

### Fines re-cycling system

| Component              | Specification                                                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>System Type</b>     | Fines recycling system with screw conveyor for conveying entrapped fines.                                                             |
| <b>Functionality</b>   | The system conveys fines from the bag house to the Hot aggregate elevator, which feeds the filler weigh hopper atop the mixing tower. |
| <b>Connected Power</b> | 10 HP (7.5 kW)                                                                                                                        |



## Hot elevator

| Specification       | Details                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------|
| Type                | Centrifugal discharge type bucket elevator                                                                         |
| Capacity            | 200 TPH (Tons Per Hour)                                                                                            |
| Connected Power     | 1 x 20 HP (15 Kw)                                                                                                  |
| Motor Power         | 20 HP (15 Kw) [Note- This specification might be changed if there is a Hot storage silo is included in the scope]. |
| Design              | Totally enclosed, vertically mounted                                                                               |
| Discharge Mechanism | Centrifugal discharge                                                                                              |
| Included Components | - Wear plates - Electrical/mechanical drives - Electrical wiring                                                   |
| Safety Feature      | Rollback arresting device                                                                                          |

## Vibrating Screen



| Specification                      | Details                                                                                                                                                                                                     |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Type</b>                        | 5 decks, inclined circular vibration type                                                                                                                                                                   |
| <b>Capacity</b>                    | 180 TPH (Tons Per Hour)                                                                                                                                                                                     |
| <b>Dimensions (Width x Length)</b> | 1500mm wide x 4275mm long                                                                                                                                                                                   |
| <b>Motor</b>                       | 2 x 8 HP (6.2 Kw) vibrating motors                                                                                                                                                                          |
| <b>Screen Sieve Sizes</b>          | <ul style="list-style-type: none"> <li>- 3 x 4 mm</li> <li>- 7 x 7 mm</li> <li>- 19 x 19 mm</li> <li>- 25 x 25 mm</li> <li>- 40 x 40 mm</li> </ul> (Note : This specs are customizable as per requirements) |
| <b>Included Components</b>         | <ul style="list-style-type: none"> <li>- One set of sieves with specified sizes</li> <li>- Duplex Spring absorbers</li> <li>- By-pass system</li> </ul>                                                     |
| <b>Design Features</b>             | <ul style="list-style-type: none"> <li>- Inclined circulating, vibrating type</li> <li>- Circular motion to minimize clogging</li> </ul>                                                                    |
| <b>Sieve Replacement</b>           | <ul style="list-style-type: none"> <li>- Lower two deck sieves hooked at longitudinal ends for easy replacement</li> <li>- No need to remove upper deck sieves for replacement</li> </ul>                   |
| <b>Maintenance Access</b>          | Wide platform around screen for easy and safe exchange or adjustment of sieves and maintenance                                                                                                              |

### Hot aggregates bin



| Specification                  | Details                                                                                                                                                                                |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Number of Compartments</b>  | 5                                                                                                                                                                                      |
| <b>Total Storage Capacity</b>  | 20.0 m <sup>3</sup>                                                                                                                                                                    |
| <b>Included Features</b>       | <ul style="list-style-type: none"> <li>- Overflow and oversize rejection chute</li> <li>- Pneumatic controlled cut-off gates</li> <li>- Sampling device on each compartment</li> </ul> |
| <b>Level Indicators</b>        | One-point bin level indicator in each compartment                                                                                                                                      |
| <b>Temperature Measurement</b> | Thermocouple-type temperature sensor in bin no. 1                                                                                                                                      |

## Weighing system

| Specification                 | Details                                                                                                                                                                                   |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Weighing System Type</b>   | Automatic accumulating and individual weighing type                                                                                                                                       |
| <b>Aggregate Weigh Hopper</b> | <ul style="list-style-type: none"> <li>- Accumulating type</li> <li>- Capacity: 3,700 kg (2.3 m<sup>3</sup>)</li> <li>- 2 discharge gates</li> <li>- Supported by 4 load cells</li> </ul> |
| <b>Filler Weigh Hopper</b>    | <ul style="list-style-type: none"> <li>- Individual weighing type</li> <li>- Capacity: 225 kg (0.28 m<sup>3</sup>)</li> </ul>                                                             |

|                             |                                                                                                                                                                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | <ul style="list-style-type: none"> <li>- Butterfly valve</li> <li>- Supported by 3 load cells</li> </ul>                                                                                      |
| <b>Asphalt Weigh Hopper</b> | <ul style="list-style-type: none"> <li>- Individual weighing type</li> <li>- Capacity: 270 kg (0.27 m<sup>3</sup>)</li> <li>- Butterfly valve</li> <li>- Supported by 3 load cells</li> </ul> |



### Pug mill type mixer unit

| Specification                 | Details                                                                   |
|-------------------------------|---------------------------------------------------------------------------|
| <b>Type</b>                   | Twin shaft jacketed pug mill                                              |
| <b>Mixing Capacity</b>        | 2250 kg per batch                                                         |
| <b>Dimensions (L x W x H)</b> | 2500 x 1,530 x 970 mm                                                     |
| <b>Connected Power</b>        | 2 x 40 HP (30 Kw)                                                         |
| <b>Construction Features</b>  | - Welded steel shell- Lined with replaceable wear-resistant spiral liners |
| <b>Mixing Mechanism</b>       | - Specially designed arms and tips for a homogeneous mix in a short time  |
| <b>Discharge Gate</b>         | - Wide opening air-operated gate- Operated by 2 cylinders                 |
| <b>Drive System</b>           | Shaft-mounted gear box                                                    |

|                        |                               |
|------------------------|-------------------------------|
| <b>Bearings</b>        | Spherical bearings            |
| <b>Truck Clearance</b> | 3,480 mm high x 3,700 mm wide |

## Fines / Filler Feeding System



- 20M3 filler silo with bucket elevator 5 HP (3.75 KW) motor for feeding filler into filler weighing hopper.  
Reclaim filler screw to feed reclaim dust from bag filter 5HP

## Asphalt Pumps

A 10 HP (7.5 kW ) 450 Lpm Jacketed, gear type asphalt transfer pump is provided for feeding asphalt from the storage tank to the asphalt weigh hopper



|                 |                            |
|-----------------|----------------------------|
| <b>Type</b>     | Hot oil jacketed gear pump |
| <b>Pressure</b> | 2 kgf/cm <sup>2</sup>      |
| <b>Capacity</b> | 450 L/min                  |
| <b>Motor</b>    | 10 HP (7.5.Kw)             |

### Bitumen Storage And Heating System

| Specification          | Details                                               |
|------------------------|-------------------------------------------------------|
| <b>Number of Tanks</b> | 2                                                     |
| <b>Capacity (Each)</b> | 50,000 liters                                         |
| <b>Design</b>          | Cylindrical, horizontal, fully insulated, and cladded |
| <b>Features</b>        | - Internal heat transfer coils- Temperature indicator |



|                                   |                                                                                                                 |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Thermic Oil Heating System</b> |                                                                                                                 |
| <b>Type</b>                       | Normal pressure, oil heating horizontal tubular boiler                                                          |
| <b>Capacity</b>                   | 500,000 kcal/h input                                                                                            |
| <b>Fuel</b>                       | High-speed diesel fuel or ASTM D-396 No. 4 fuel oil                                                             |
| <b>Burner</b>                     |                                                                                                                 |
| - <b>Model</b>                    | FCMP/LO/S/60                                                                                                    |
| - <b>Type</b>                     | High-pressure mechanical atomizing, gun type oil burner                                                         |
| - <b>Capacity</b>                 | 18 to 60 L/h                                                                                                    |
| - <b>Control</b>                  | Automatic ignition with high-low control                                                                        |
| - <b>Safety Controls</b>          | - Flame eye- Low-level cut-out float switch- Pressure switch- High-temperature cut-out thermostat               |
| <b>Piping and Controls</b>        |                                                                                                                 |
| - <b>Hot Oil Transfer Piping</b>  | Total system with necessary control valves                                                                      |
| - <b>Bitumen Piping</b>           | Jacketed piping with process valves and fittings for manual tank selection and single bitumen delivery pipeline |
| <b>Hot Oil Transfer Pump</b>      |                                                                                                                 |
| - <b>Type</b>                     | Centrifugal pump                                                                                                |
| - <b>Capacity</b>                 | 30 m <sup>3</sup> /hour                                                                                         |
| - <b>Head</b>                     | 55 m                                                                                                            |
| - <b>Connected Power</b>          | 10 HP (7.5 Kw)                                                                                                  |
| <b>Auxiliary Power</b>            | 1 HP (0.75Kw)                                                                                                   |

## Fuel Tank

24 000 Ltrs capacity, vertical, cylindrical tank for LDO

## Control Cabins



### Comprising Of Operator's Control Cabin

2500 mm wide x 3000 mm long, Fully Air-conditioned operator's control cabin complete with weather proof exterior and fitted with tinted toughened glass windows and flush entrance door with lock.

Interconnecting

wiring (related to all the above sections) designed to suit the positioning of the unit within 10 m radius of the mixing unit.

### Electrical Power Control Console

Complete Motor control console for direct online starting and for star delta starting on necessary devices.

### Distribution Switch Board

- Completely enclosed steel cubicle with push buttons starting of motors on star delta starters.
- Magnetic switches with overload thermal relay for each individual motor.
- Safety Interlocks and protection devices.
- Meters for online current voltage measurements for high capacity motors.
- Emergency stop button.
- Short circuit protection with switch fuse unit.
- Inverter Drives for control of cold aggregates bin feeders

### Temperature Indication

- 0 -300 Deg C, Electronic multi point scanner for measuring:
- Aggregates temperature measuring in Hot Bin no. 1.
- Asphalt feeding temperature.
- Bag filter inlet gas temperature.
- Bitumen tank temperature. (tank # - 1 & 2)

### Sequence Control panel

- Programmable Logic Control (PLC) with man machine interface.
- Fully automatic electronic solid state sequence control.
- Operator override for manual or semi automatic control.
- Timer controls for both wet and dry cycle.
- Pilot lamps for indication of sequence currently operated.
- Signals and alarms.
- Storage of Fifty different type of mix proportions.
- Automatic free-fall compensation.
- Direct dialing of mix design by percentage.
- Preset batch counter controls the number of batches for a truck-full load.
- Provision for printing operating data on a day to day basis, like - Mix proportions, Batch weight, Total no. of batches, Subtotal, gross total etc.

### **Dryer & Burner control.**

- Hot aggregates temperature at dryer discharge end.
- Remote Burner flame synchronization with hot aggregates temperature.
- Safety interlock, in case of high temperature.

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## OPTIONAL ACCESSORIES TO MAKE YOUR PLANT AFFORDABLE

| Accessory                                  | Functionality                                                                         | Technical Details                                                                                                                                                                                     |
|--------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cold RAP (up to 40%)</b>                | Allows the use of recycled asphalt materials in the mix without preheating.           | <ul style="list-style-type: none"> <li>- RAP typically contains 3-6% residual bitumen and aggregates.</li> <li>- Efficient blending ensures mechanical and thermal stability.</li> </ul>              |
| <b>Hot RAP – HRC (up to 60%)</b>           | Incorporates pre-heated recycled asphalt materials into the mix.                      | <ul style="list-style-type: none"> <li>- Requires precise temperature control to prevent bitumen oxidation.</li> <li>- Can maintain the binder's rheological properties.</li> </ul>                   |
| <b>Air Atomized Burner</b>                 | Utilizes compressed air to atomize fuel for the drying drum.                          | <ul style="list-style-type: none"> <li>- Atomization improves fuel droplet surface area for combustion.</li> <li>- Operates with high thermal efficiency (~90%).</li> </ul>                           |
| <b>Coal Burner</b>                         | Uses coal as a fuel source for the drying drum.                                       | <ul style="list-style-type: none"> <li>- Requires pulverized coal for uniform combustion.</li> <li>- Emission control measures like scrubbers or ESPs are recommended.</li> </ul>                     |
| <b>Gas Burner (for Bitumen Tank)</b>       | Uses LPG or natural gas to heat the bitumen tank.                                     | <ul style="list-style-type: none"> <li>- Gas combustion achieves &gt;95% efficiency.</li> <li>- Reduced NO<sub>x</sub> and SO<sub>x</sub> emissions compared to liquid fuels.</li> </ul>              |
| <b>CNG Burner (for Thermic Oil Heater)</b> | Employs compressed natural gas to heat thermic oil used for asphalt plant operations. | <ul style="list-style-type: none"> <li>- Methane (CH<sub>4</sub>) in CNG burns at a high calorific value (~50 MJ/kg).</li> <li>- Produces 30-40% fewer emissions than coal or oil burners.</li> </ul> |

|                                                            |                                                                    |                                                                                                                                                                                               |
|------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CNG Burner (for Drum Burner of Batch Asphalt Plant)</b> | Heats the aggregate in batch asphalt plants using CNG.             | <ul style="list-style-type: none"> <li>- Requires proper burner calibration for optimal mixing of gas and air.</li> <li>- Generates consistent flame temperature (~1,900°C).</li> </ul>       |
| <b>Thermic Oil Heater with Coal/Wood Fire Burner</b>       | Provides heating using thermic oil heated by burning coal or wood. | <ul style="list-style-type: none"> <li>- Efficient thermic fluid systems achieve 85-90% heat transfer efficiency.</li> <li>- Requires a well-insulated heating coil for longevity.</li> </ul> |

### **OTHER ANCILLIARY ITEM**

| <b>Ancillary Item</b>                          | <b>Functionality</b>                                                                                            | <b>Technical Details</b>                                                                                                                                                                            |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Foam Bitumen                                   | Creates foamed bitumen by injecting water into hot bitumen, expanding its volume and improving mix workability. | <ul style="list-style-type: none"> <li>- Bitumen expands up to 10-15 times its volume.</li> <li>- Optimal water content (~2%) and temperature (~160-180°C) critical for foam stability.</li> </ul>  |
| Stone Mastic Asphalt (SMA)                     | A high-performance asphalt mix with a stone skeleton and bitumen-mastic matrix for heavy-duty surfaces.         | - SMA typically includes stabilizing additives like cellulose or polymer fibers to prevent bitumen drain-off.                                                                                       |
| Liquid Additive                                | Enhances bitumen properties for better adhesion, flexibility, and temperature resistance.                       | - Additives like amines or polymers chemically modify bitumen to enhance adhesion and aging resistance.                                                                                             |
| Inline Silo (25-300 Tons, 1-3 Compartments)    | Storage solution for aggregates or finished products, ensuring segregation and availability.                    | <ul style="list-style-type: none"> <li>- Designed for easy loading and retrieval, with compartments preventing material mixing.</li> <li>- Load cells ensure accurate material tracking.</li> </ul> |
| Hoist for Easy Maintenance (Elevator & Screen) | Mechanical lifting system for accessing and maintaining critical plant components.                              | - Typically uses electric or hydraulic systems with lifting capacities suited for heavy plant components.                                                                                           |
| Dust Conditioner                               | Mixes water with collected dust to form a slurry for safe disposal or recycling.                                | <ul style="list-style-type: none"> <li>- Uses water spray systems for consistent wetting.</li> <li>- Recycled water systems enhance sustainability.</li> </ul>                                      |

